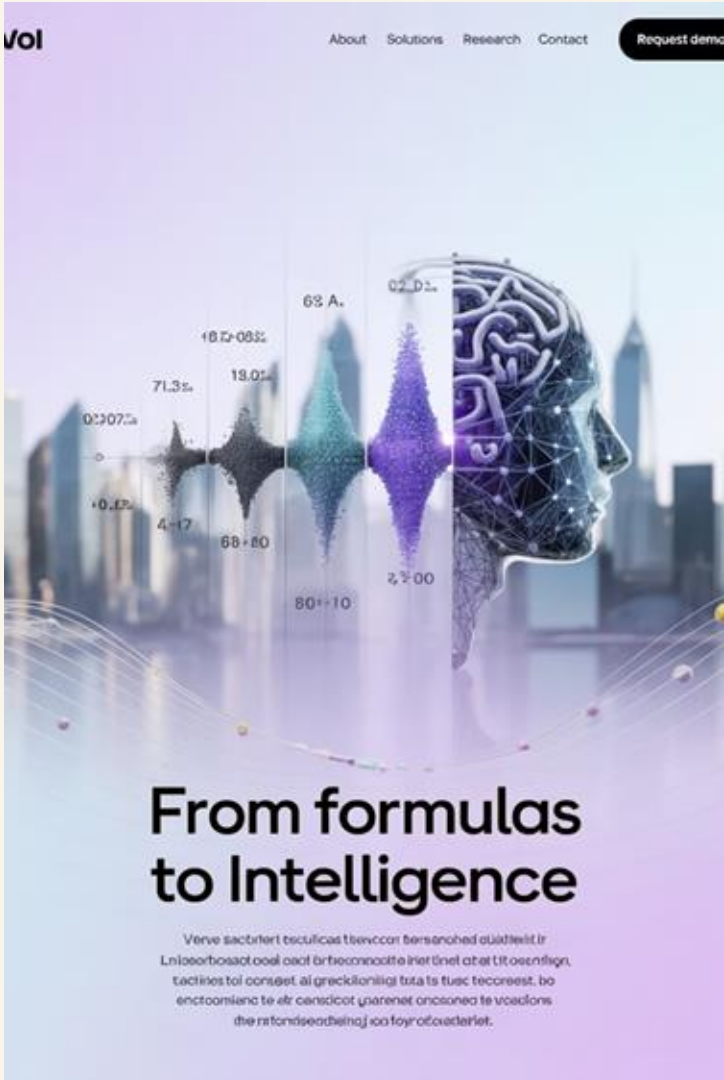
Applied Finance Projects

From mathematical models to generative AI

Andy Li, PhD, CFA, FRM



Guest Speaker’s bio

Alex Zhang has over 15 years of experience in structured credit and quantitative investing. Most recently, he served as Portfolio Manager and Head of Structured Credit at Squarepoint Capital in New York. Since joining the firm in 2020, Alex built and led the structured credit platform from inception, overseeing investments across CLOs, non-agency RMBS, CMBS, ABS, and SRT strategies. He was responsible for portfolio management and trading, while also leading the development of quantitative research, pricing models, and the analytical infrastructure supporting the platform.

Prior to Squarepoint, Alex was a Structured Credit Trader at Guggenheim Partners. Earlier in his career, he held portfolio management and trading roles at several hedge funds, focusing on structured products and quantitative investment strategies.

Alex holds a Master of Financial Engineering from the University of California, Berkeley (Haas School of Business), a Master of Science in Mathematics from Northwestern University, and a Bachelor of Science in Mathematics from the University of Science and Technology of China.

Questions for Alex

1. What differentiates a good quant in structured credit from someone you would trust to run risk as a PM?

1. How do you handle model assumptions breaking down in stressed markets?

1. How do you balance quant signals vs. trader judgment in decisions?

1. How do you manage liquidity and tail risk in illiquid products?

1. If you rebuilt your platform today, what would you do differently?

1. How important is knowing AI techniques in this field?

1. How can AI engineering such as LLM and Machine learning lower the barrier to entry or be more marketable in your sector?

1. Could you share with us some interesting experiences you have had in your career?

Applied Finance Project

With the Applied Finance Project, you will:

- Apply your newly acquired **quantitative skills, tools, and AI techniques**
- Work on a **real-world financial problem** inspired by industry use cases
- Build a **production-style system** (RAG / Agent / Financial AI application)
- Translate theory into **practical, decision-relevant insights**

What You Will Gain

- **Hands-on experience** with modern AI + finance workflows
- A **portfolio project** for internships / full-time roles
- Exposure to **real industry problems** (equity research, risk, filings, etc.)
- Stronger **storytelling for interviews**

Team-Based Learning

- Work in teams of **3–4 students**
- Collaborate across:
  - Data & ingestion
  - Model / RAG pipeline
  - Agent & reasoning
  - UI / product

Examples

AI Earnings Call Analyst: Project Overview

Theme: RAG for financial transcript analysis

⏱️ THE CHALLENGE & SOLUTION ❓

Problem
Earnings call transcripts are long and repetitive; analysts need fast answers on guidance, risks, tone, and business priorities.

Goal 🎯🤖
Build an AI assistant to answer questions from transcripts and generate a short analyst-style summary.

⚙️ PROJECT EXECUTION 💻

What Students Build

- Ingest 3–5 quarters of transcripts for 3–5 companies
- Create a retrieval pipeline
- Support Q&A over transcripts
- Generate a concise "what changed this quarter" summary.

Data Sources 🗄️☁️

- Company investor relations websites
- Public earnings call transcripts
- Optional: Earnings press releases

💬 EXAMPLE QUESTIONS

- What were the main risks management mentioned this quarter?
- How did guidance change versus last quarter?
- What themes appeared repeatedly across the last 3 calls?
- Did management sound more optimistic or cautious?

AI Portfolio Risk Monitor Agent

Theme: agent + multi-source retrieval for portfolio monitoring

PROBLEM & GOAL

Problem

Investors holding 10–20 names cannot manually monitor all news, filings, macro updates, and exposures daily.

Goal

Build an agent that monitors a sample portfolio and produces a daily/weekly risk memo.

PROJECT EXECUTION: WHAT STUDENTS BUILD

Take a portfolio of 5–10 stocks

↓

Retrieve recent company news, selected SEC filings, optional market indicators

↓

Use an LLM agent to summarize key risks

↓

Produce a portfolio-level risk dashboard or memo.

EXAMPLE OUTPUT

- Top 3 portfolio risks this week
- Stock-level risk flags
- Common themes across holdings
- Suggested areas for analyst review

EXAMPLE QUESTIONS

- What are the biggest current risks in this portfolio?
- Which names had the most negative developments this week?
- Are there common risk themes across semiconductor holdings?
- Which holdings have new litigation, regulatory, or guidance risk?

DATA

- market/news APIs or curated articles
- SEC filings
- portfolio list
- optional macro data

SEC Filing AI Research Assistant

Theme: RAG for long-form regulatory filings

Problem

10-Ks and 10-Qs contain critical information, but they are too long for efficient manual review.

Goal

Build an AI research assistant that answers analyst-style questions from SEC filings and cites the relevant filing sections.

What students build

- download filings for 3–5 public companies
- parse and chunk the filings
- build a filing QA assistant
- return answers with source excerpts or section references

Data

- SEC EDGAR 10-K and 10-Q filings
- optional proxy statements or 8-Ks

Example questions

- What are the company's main business segments?
- What risk factors are most relevant to this company?
- What changed in MD&A from last year?
- What does the company say about supply chain, competition, or regulation?

PROJECT EXECUTION: WHAT STUDENTS BUILD



download filings for 3–5 public companies

build a filing QA assistant

return answers with source excerpts or section references

DATA

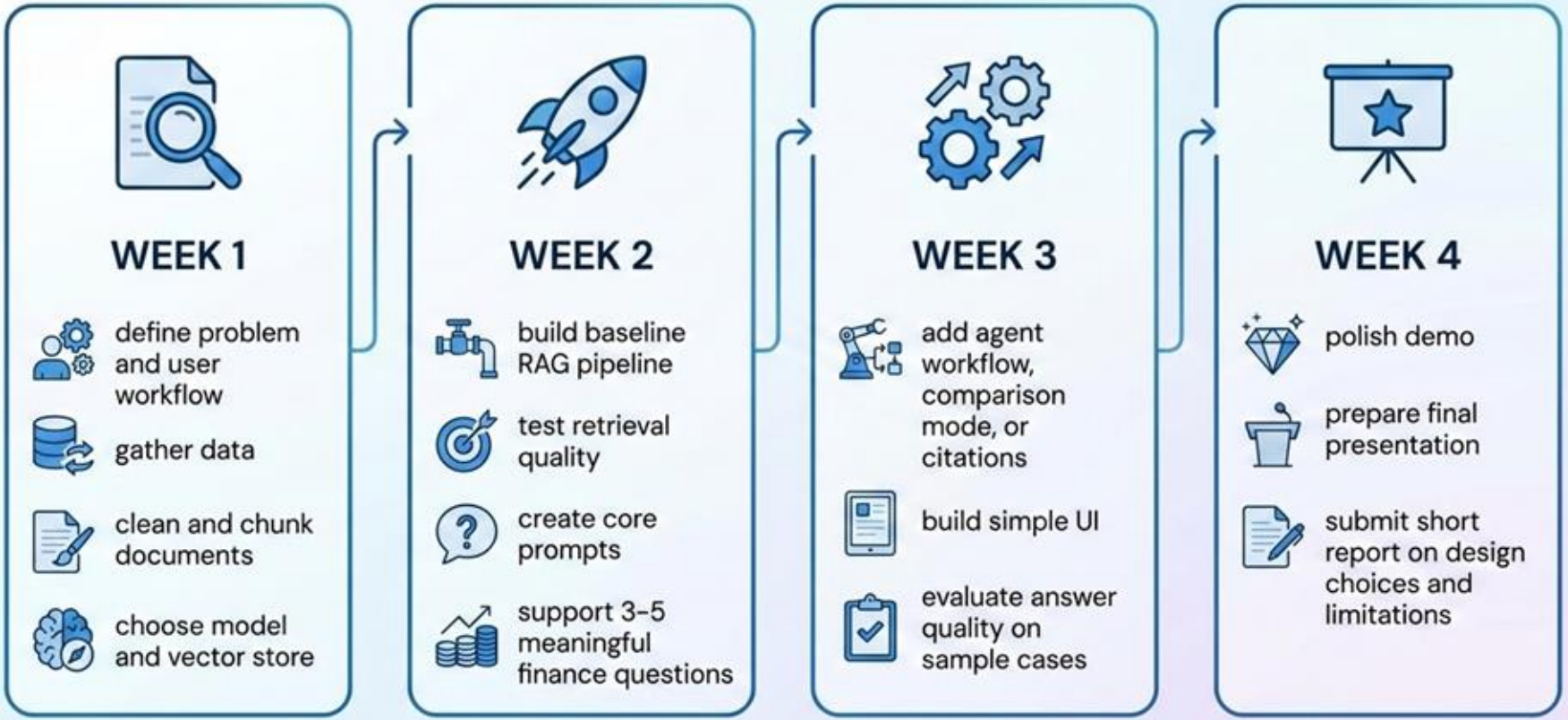
SEC EDGAR 10-K and 10-Q filings
optional proxy statements or 8-Ks

Recommended deliverables

Each team should submit:

- working demo
- GitHub repo
- short presentation
- 2–3 page writeup covering:
  - business problem
  - architecture
  - dataset
  - evaluation
  - limitations
  - → next steps

Suggested 3–4 Week Structure



https://morning-edge.vercel.app/